

Class09: Candy Mini Project

Madina Khorami (A18555185)

Table of contents

Background	1
Data Import	1
Exploratory Analysis	4
Overall Candy Ranking	7
Taking a look at pricepercent	16
Exploring the Correlation	22
Principal Component Analysis	23
Summary	25

Background

Today we are taking a detour to analyze a fun and dataset (that we have more intrinsic insight into) with the most useful method we have learned thus far - Principal Component Analysis (PCA).

Data Import

The data is all related to Halloween can and is from 538 websites.

```
candy <- read.csv("candy-data.csv", row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0

Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0
	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

We can access our favorite candy type and find it's winpercent by just using the `winpercent()` or from the `dplyr` package.

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["M&M's", ]$winpercent
```

```
[1] 66.57458
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="M&M's") |>
  select(winpercent)
```

```
      winpercent
M&M's    66.57458
```

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes. The Winpercent column has values that are beyond 0-1 scale that other columns have.

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset. Let’s install this package and try it on our candy data.

```
## This way we just used one function from the package not downloading the whole skimr package
skimr::skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12

Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice-wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q7. What do you think a zero and one represent for the candy\$chocolate column?

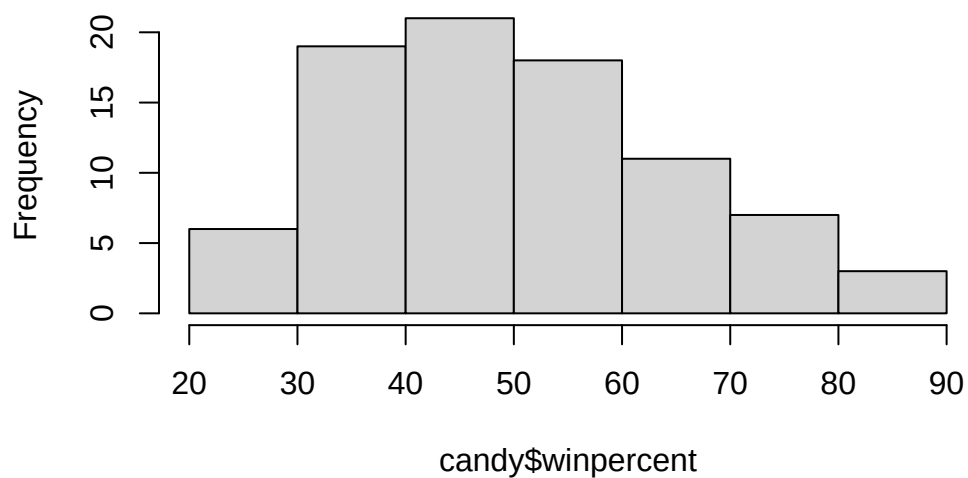
In the Chocolate column the 0 represents FALSE meaning that the candy does not contain chocolate, and 1 represents TRUE meaning the candy contains chocolate.

Exploratory Analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

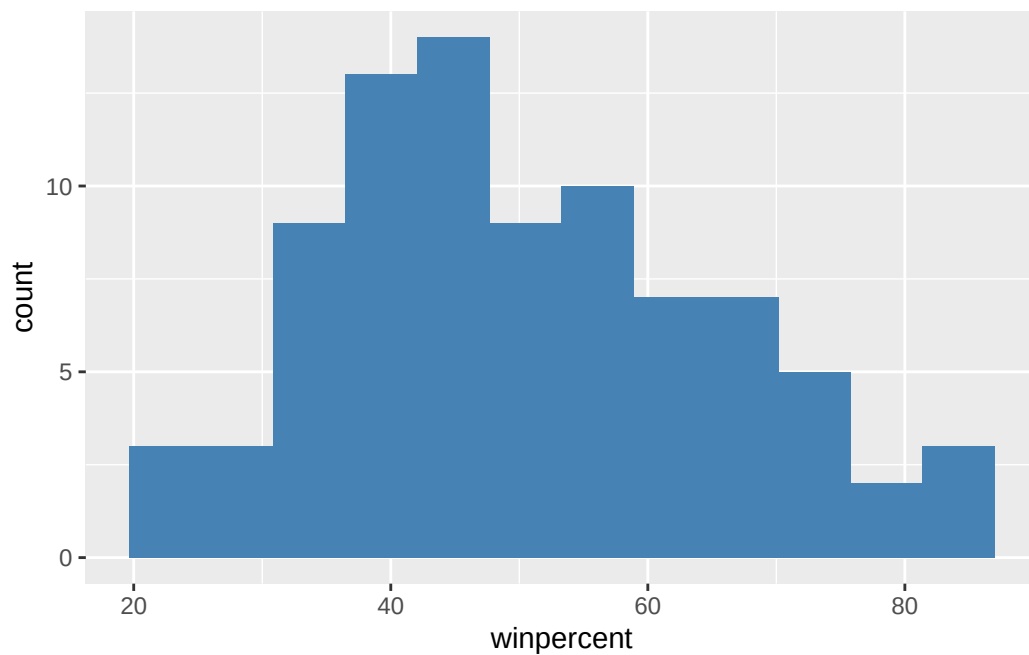
```
hist(candy$winpercent)
```

Histogram of candy\$winpercent



```
library(ggplot2)

ggplot(candy) + aes(winpercent) + geom_histogram(bins = 12, fill = "steelblue")
```



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution of the winpercent values is not symmetrical.

Q10. Is the center of the distribution above or below 50%?

The center of the distribution is above 50%.

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Based on the summary data the `median()` best tells what the center of the histogram which is 47% and that is below 50%/.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc.ind <- as.logical(candy$chocolate)
choc.candy <- candy[choc.ind, ]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
fruity.ind <- as.logical(candy$fruity)
fruity.candy <- candy[fruity.ind, ]
fruity.win <- fruity.candy$winpercent
mean(fruity.win)
```

```
[1] 44.11974
```

Based on this comparison results chocolate is more preferred than fruity candy and has a higher rank.

Q12. Is this difference statistically significant?

```
t.test(choc.win, fruity.win)
```

Welch Two Sample t-test

```
data: choc.win and fruity.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

The difference is statistically significant.

Overall Candy Ranking

Q13. What are the five least liked candy types in this set?

```
candy |>
  arrange(winpercent) |>
  select(winpercent) |>
  head(5)
```

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

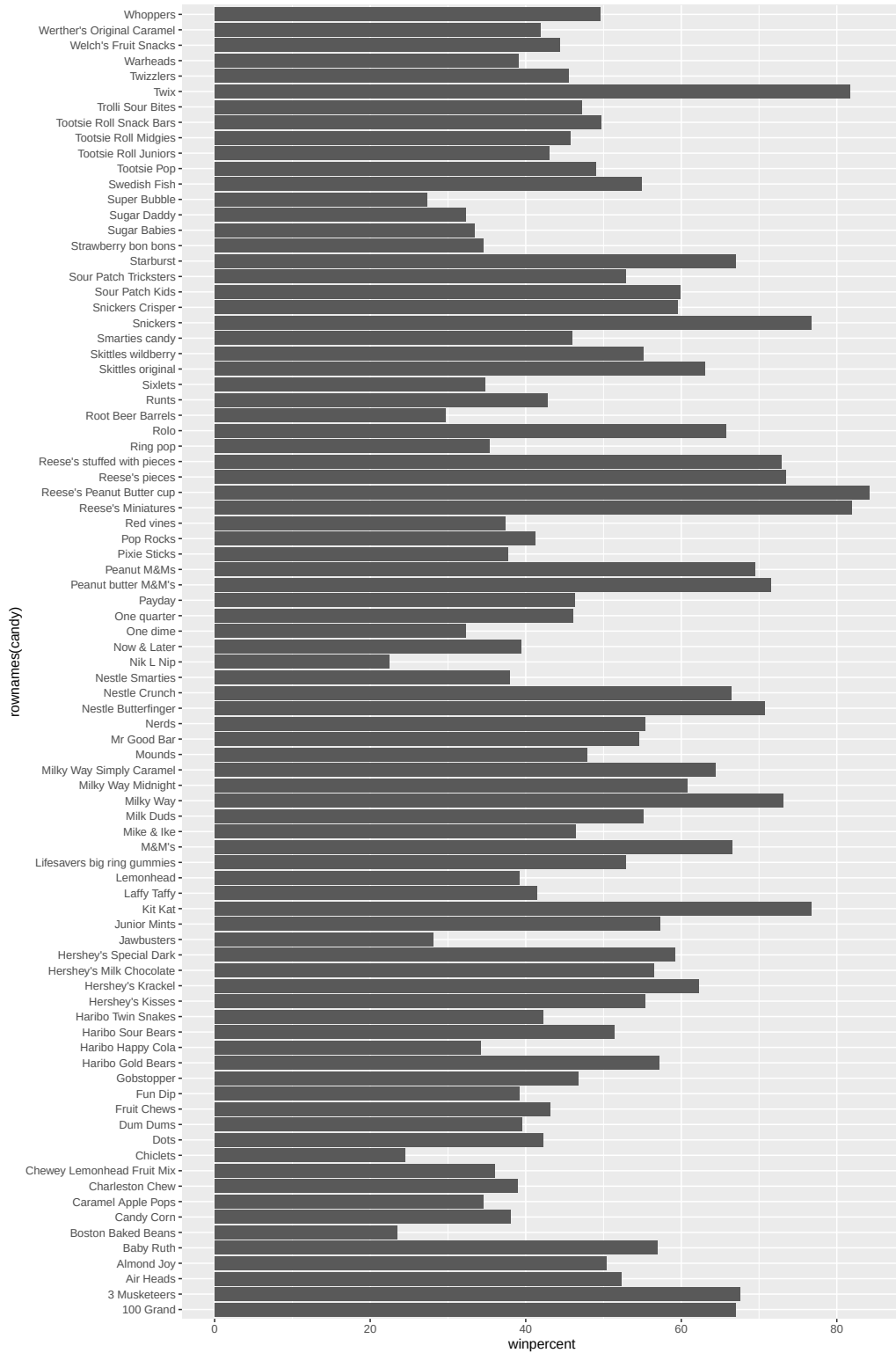
Q14. What are the top 5 all time favorite candy types out of this set?

```
candy |>
  arrange(winpercent) |>
  select(winpercent) |>
  tail(5)
```

	winpercent
Snickers	76.67378
Kit Kat	76.76860
Twix	81.64291
Reese's Miniatures	81.86626
Reese's Peanut Butter cup	84.18029

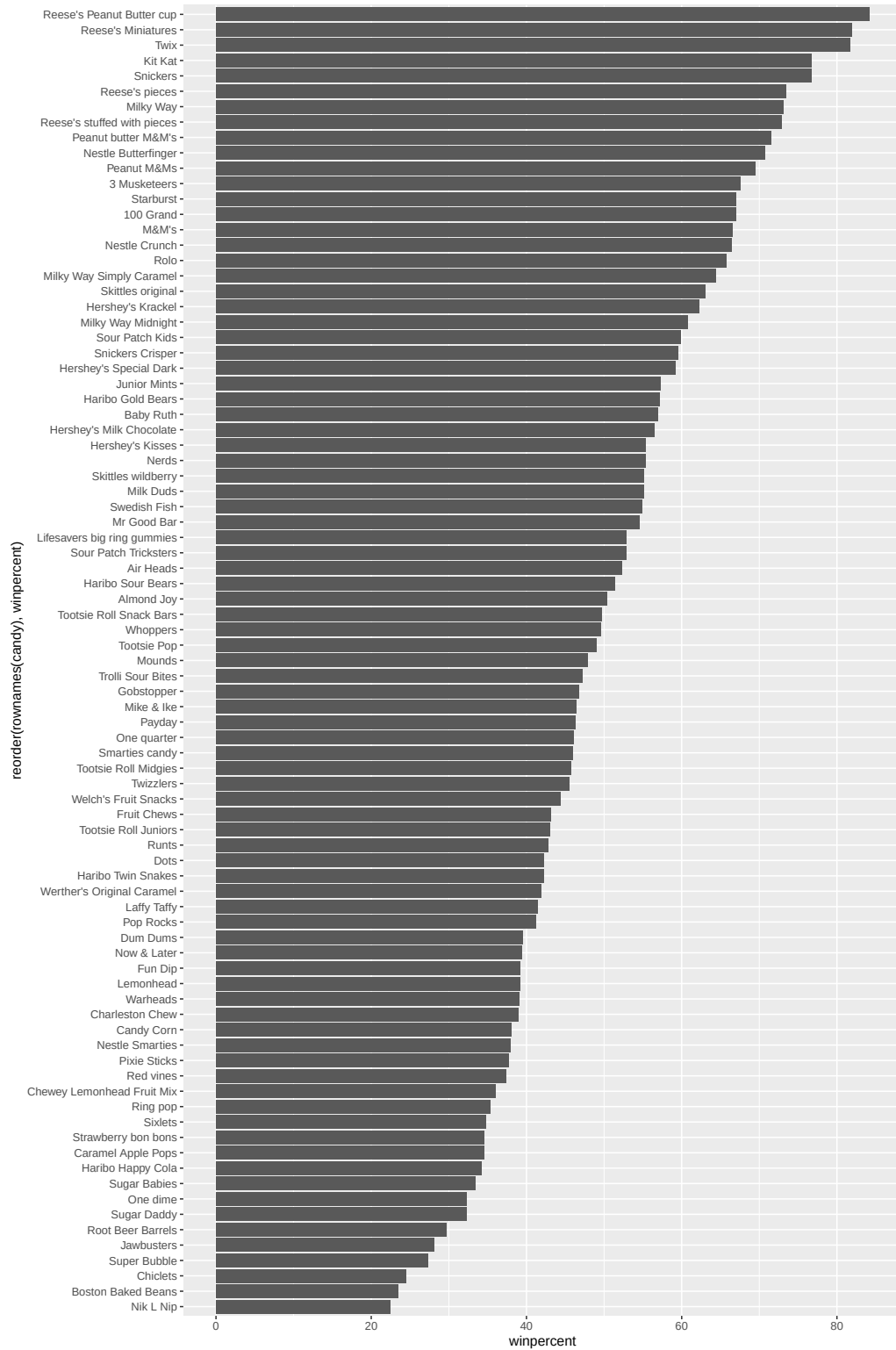
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```

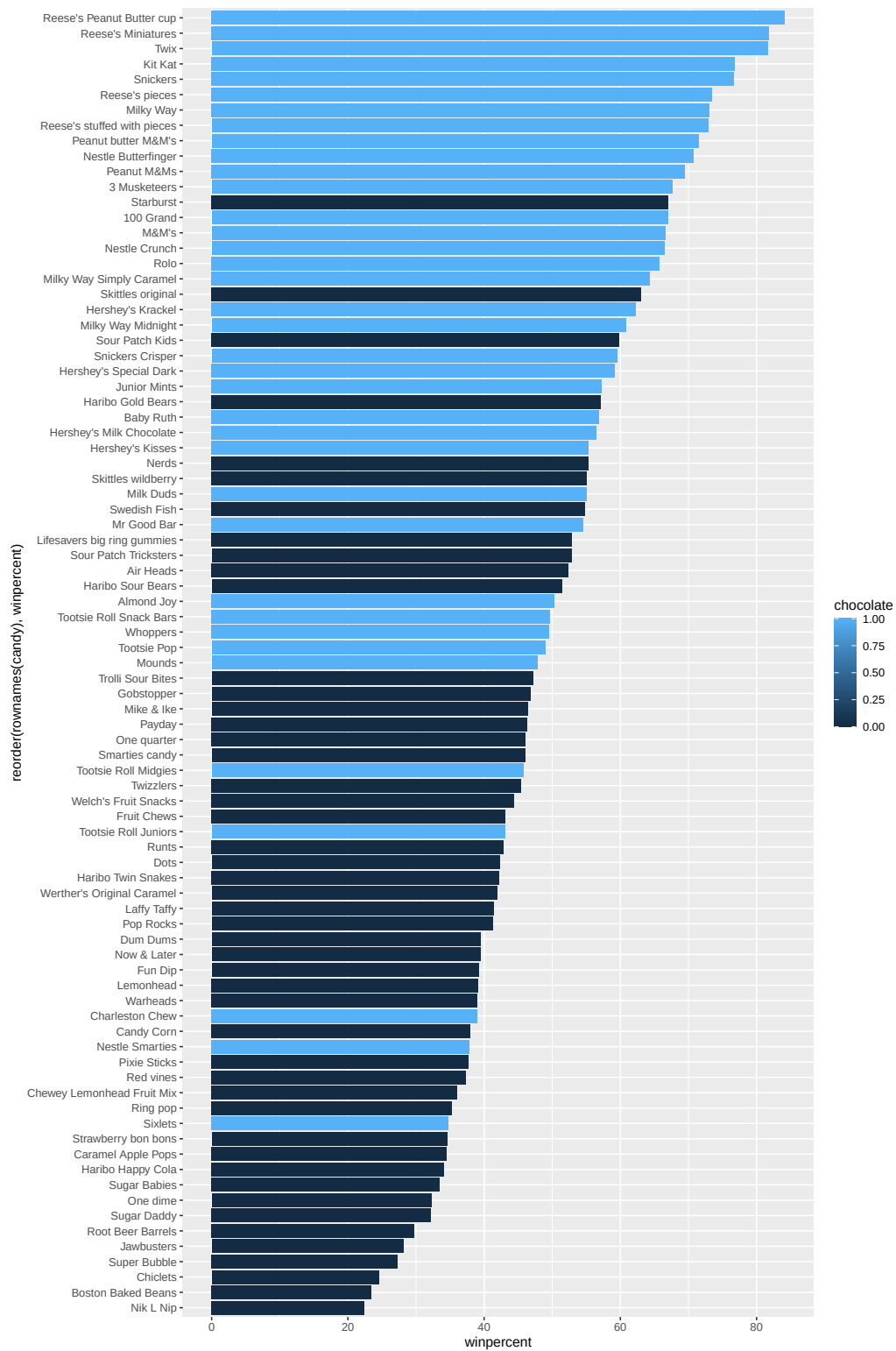


Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col()
```



```
ggplot(candy) +  
  aes(winpercent,  
       reorder(rownames(candy), winpercent),  
       fill = chocolate) +  
  geom_col()
```



Now let's try our barplot with these colors. Note that we use fill=my_cols for geom_col(). Experiment to see what happens if you use col=mycols.

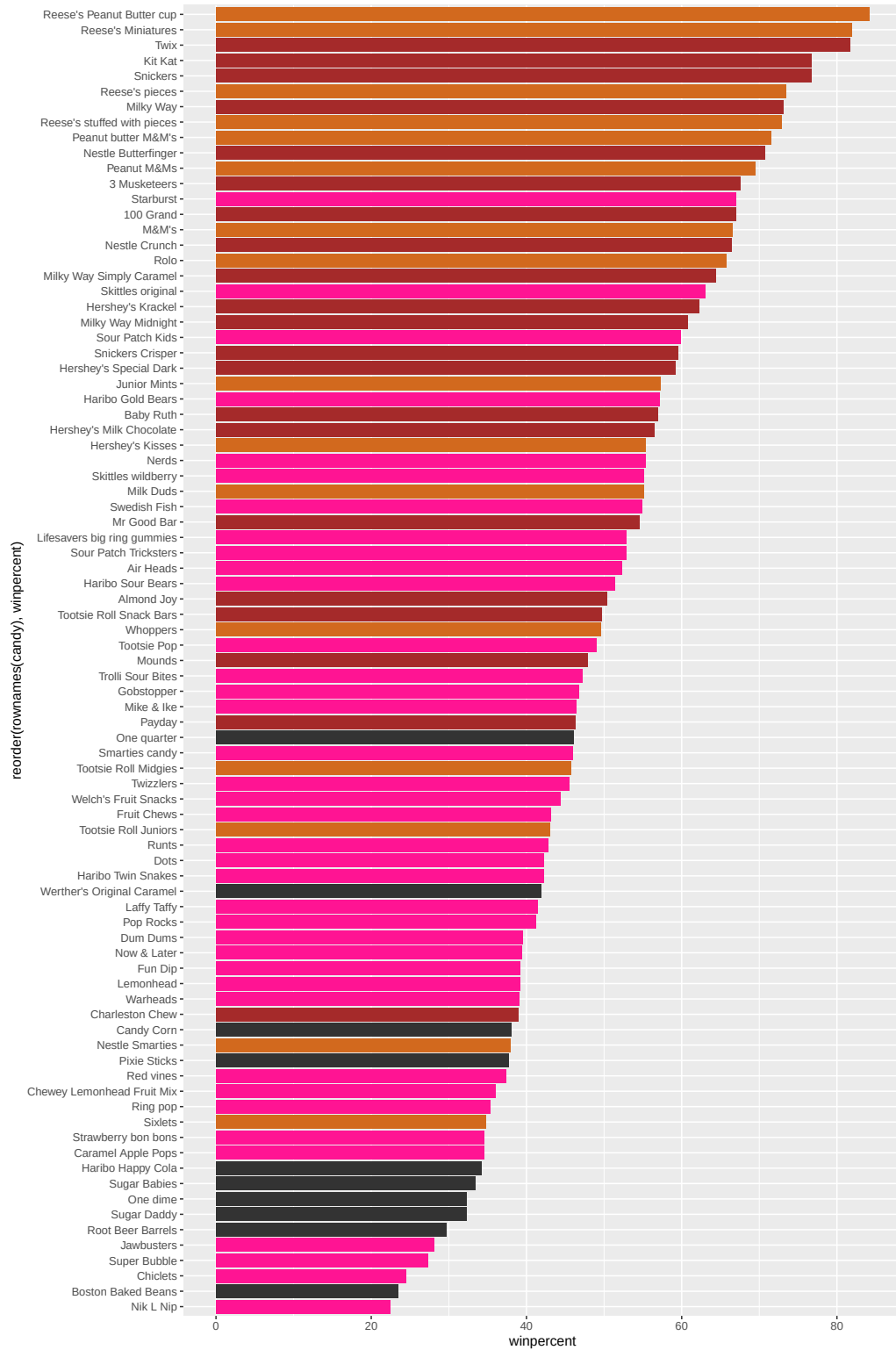
```
## rep keeps repeating the order we give to it.
mycols <- rep("gray20", nrow(candy))

## Chocolate candy
mycols[as.logical(candy$chocolate)] <- "chocolate"

## Candy bars in brown
mycols[as.logical(candy$bar)] <- "brown"

##fruity candy in pink
mycols[as.logical(candy$fruity)] <- "deeppink"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill = mycols)
```



Q17. What is the worst ranked chocolate candy?

Based on the plot the worst ranked chocolate candy is “Sixlets”.

```
rownames(candy)[candy$chocolate == 1][which.min(candy$winpercent[candy$chocolate == 1])]
```

```
[1] "Sixlets"
```

Q18. What is the best ranked fruity candy?

The best ranked fruity candy is “Starburst”.

```
rownames(candy)[candy$fruity == 1][which.max(candy$winpercent[candy$fruity == 1])]
```

```
[1] "Starburst"
```

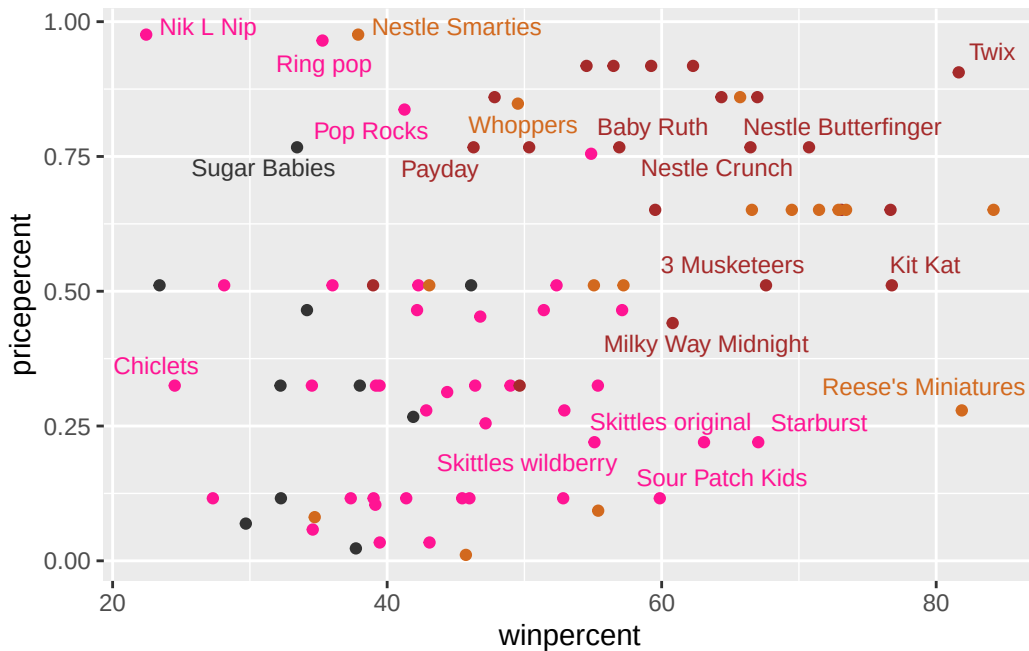
Taking a look at pricepercent

What about value for money? What is the best candy for the least money? One way to get at this would be to make a plot of **winpercent** vs the **pricepercent** variable.

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent,
      pricepercent,
      label=rownames(candy)) +
  geom_point(col=mycols) +

  # this fix the labeling overlapping with the dots
  geom_text_repel(col=mycols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

The candy that is highest ranked in terms of winpercent for the least money is chocolate.

```
min_price <- min(candy$pricepercent)
cheap_candies <- candy[candy$pricepercent == min_price, ]
best_row <- which.max(cheap_candies$winpercent)
colnames(cheap_candies)[best_row]
```

```
[1] "chocolate"
```

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

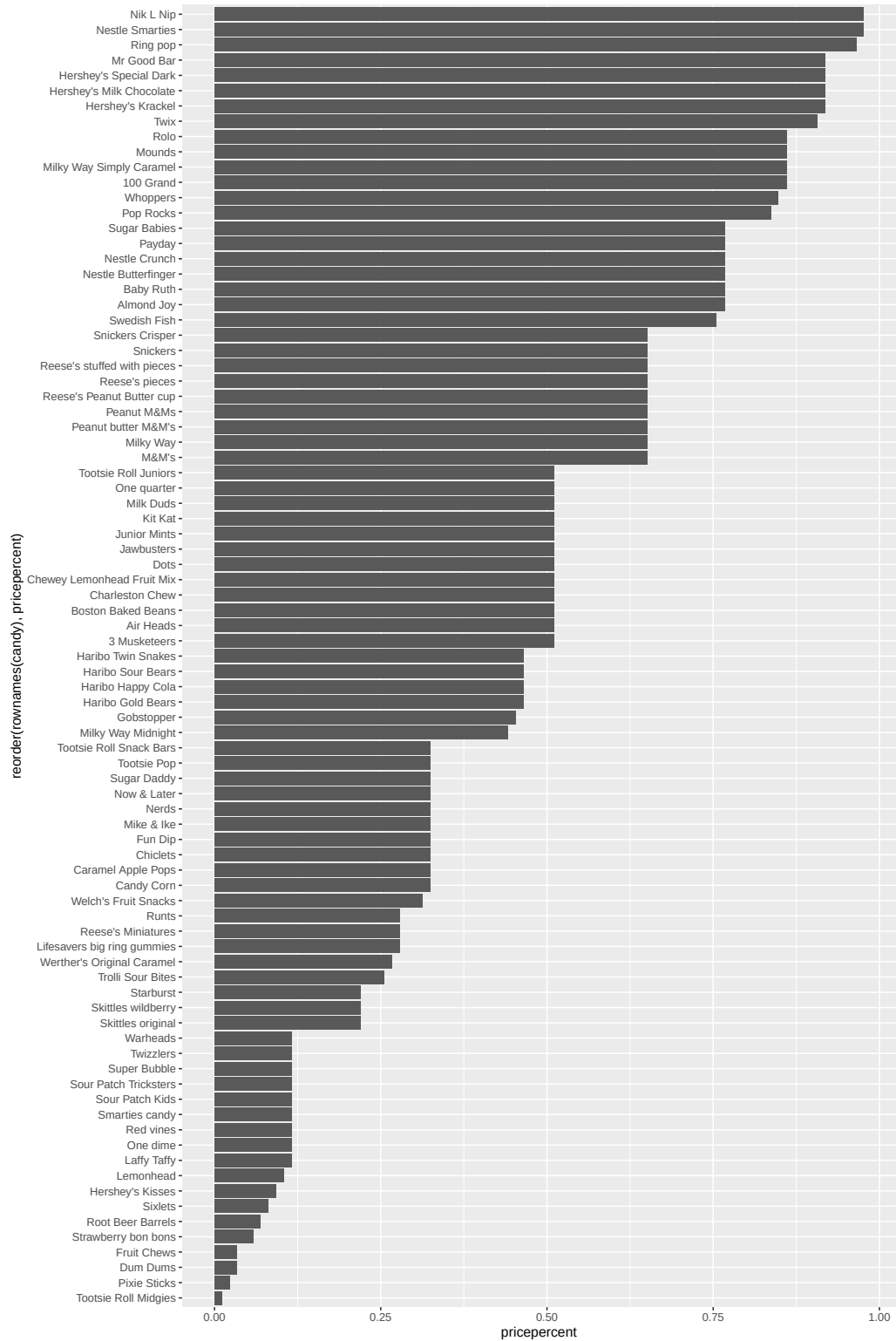
The top 5 most expensive candies are in the table below.

```
library(dplyr)
candy |>
  arrange(-pricepercent) |>
  select(pricepercent, winpercent) |>
  head(n=5)
```

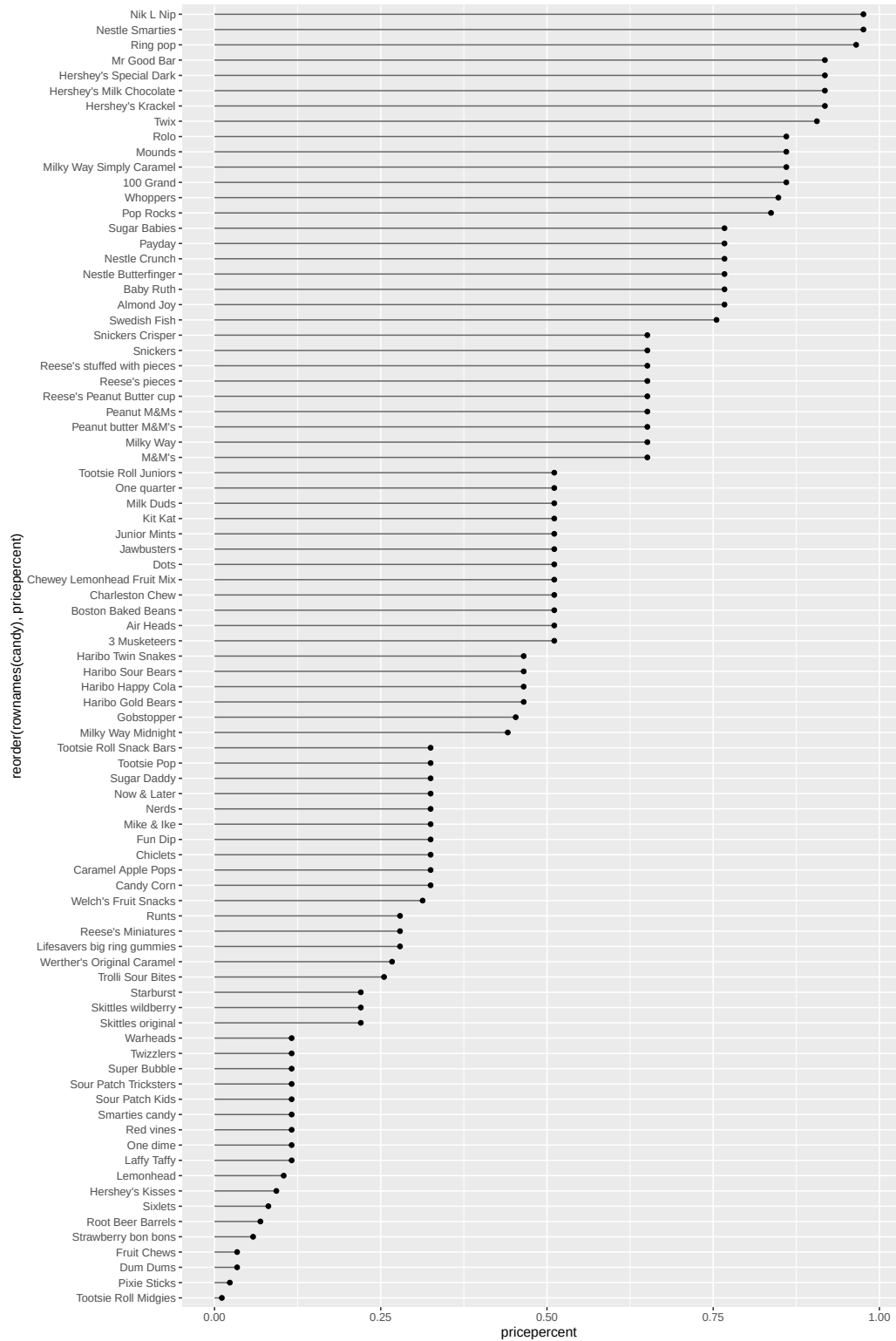
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Q21. Make a barplot again with `geom_col()` this time using `pricepercent` and then improve this step by step, first ordering the x-axis by value and finally making a so called “dot chat” or “lollipop” chart by swapping `geom_col()` for `geom_point()` + `geom_segment()`.

```
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_col()
```



```
# Make a lollipop chart of pricepercent
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent),
                    xend = 0), col="gray40") +
  geom_point()
```



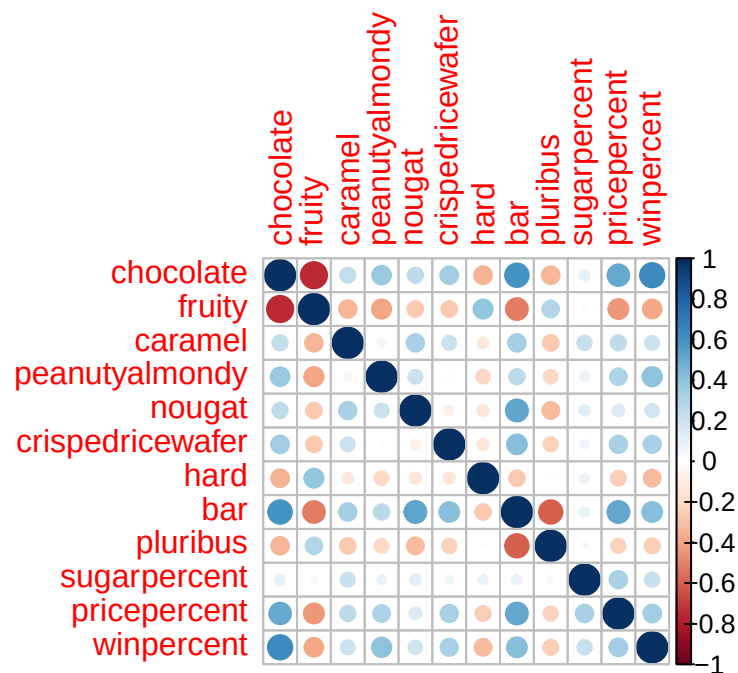
Exploring the Correlation

We can calculate the pair-wise correlation of all our columns.

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Furity and Chocolate are anti correlated variables.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

From the corrplot, the variables that will contribute most to PC1 are the ones that are strongly correlated with many others so large dark blue/red circles which are chocolate, bar, pricepercent, and winpercent because they show strong correlations and capture the main differences between candies.

Principal Component Analysis

In this case we want to be sure to set `scale=TRUE` because we have one variable `winpercent` that is on a very different scale than all other and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale. = TRUE)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

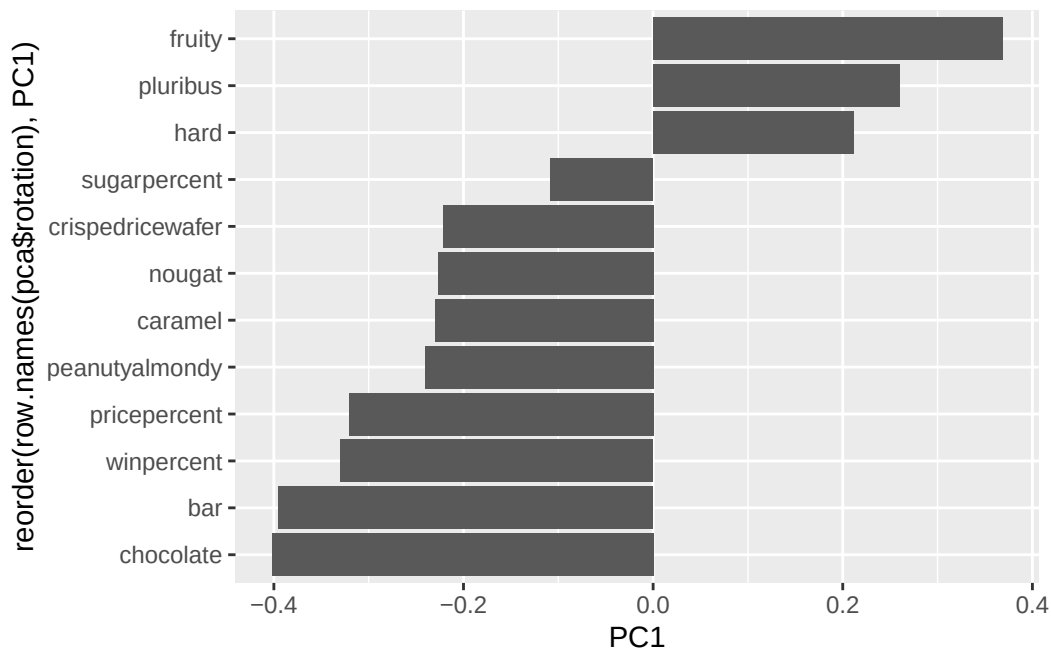
First major result figure is the “Score Plot” of PC1 vs PC2 - how different candies are related to each other on our new PC axis.

```
p <- ggplot(pca$x) +
  aes(PC1, PC2, label=row.names(pca$x)) +
  geom_point(col = mycols) +
  geom_text_repel(col = mycols, size=3.3, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
       caption="Data from 538")
```

We can make a more interactive plot by introducing the `ggplotly()`

The second major result figure from PCA is so-called “loading Plot”.

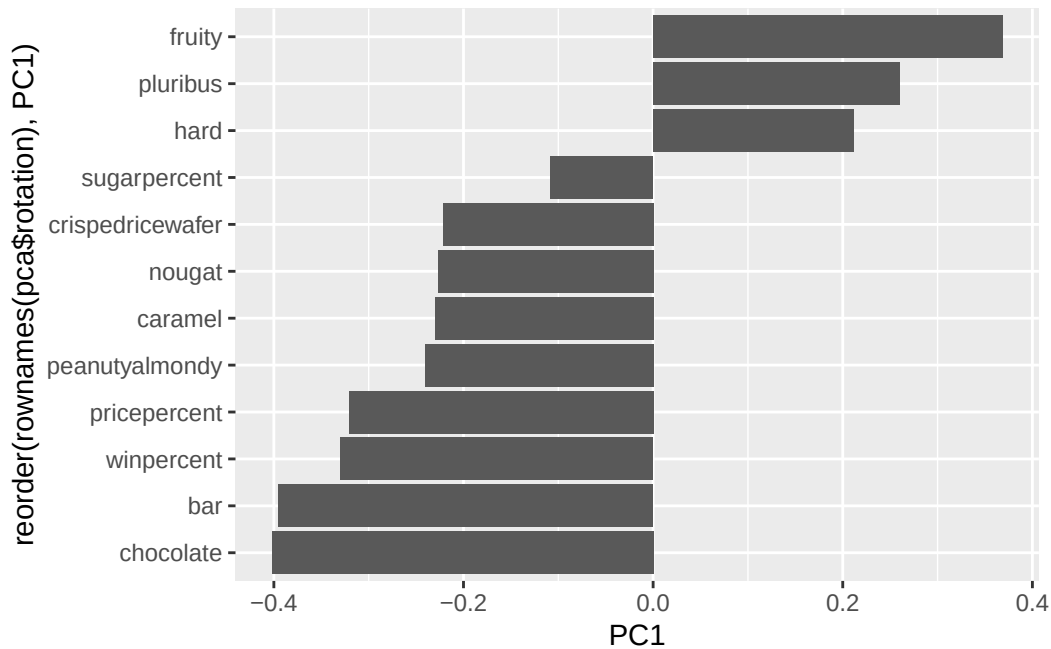
```
ggplot(pca$rotation) +
  aes(PC1,
      reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables that are strongly positive on PC1 are fruity, pluribus, and hard. This makes sense because PC1 is basically separating fruity, hard, non-chocolate candies from chocolate-type candies. On the positive side, you have fruity and hard candies, while on the negative side you have chocolate bars with things like caramel, nougat, and nuts. We already saw this pattern in the correlation plot, where chocolate and fruity were strongly opposite. It also showed up in the PCA scatter plot, where fruity candies were on one side and chocolate candies were on the other. So PC1 is clearly separating candy types, and the loading plot matches what we saw visually.

```
ggplot(pca$rotation) +
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A “winning” candy is usually chocolate-based, comes as a bar, and has things like caramel, nougat, or nuts. It’s usually not fruity, not hard, and not a pluribus type, and it tends to be a little more expensive. This makes sense because in the plots, chocolate bars were closer to the top of winpercent, while fruity and hard candies were farther away. The correlation matrix also showed that chocolate, bar, caramel, and nut features go together, while fruity and hard are kind of the opposite. The PCA showed the same pattern, with chocolate-type candies on one side and fruity/hard ones on the other, so all the results point to the same idea.